DIGISTEM · architecture

Digital Street Enumeration & Address Management System

PRESENTATION

Mobile UI
ERPNEXT INSIDE

Optimized touch forms for enumerators. Works offline, caches data via service workers.

PWA | service worker | IndexedDB

Back office web

QC dashboards, project management, GIS validation for coordinators.

responsive | Frappe UI

Executive dashboard

Real-time progress, installation tracking, budget vs actual.

charts | reports

↓HTTPS / REST↓

APP LAYER

Frappe Server (Python)

Handles all business logic, validation, workflows, and authentication.

gunicorn | redis | celery

Custom DocTypes

Street, House Unit, Session, Implementation, etc.

metadata | permissions

Workflow engine

Draft → QC → Verified → Approved. Automated assignment.

state machine

↓ORM / database queries↓

DATA

MariaDB

Transactional data: streets, units, users, sessions.

ACID | replication

File storage

House photos, shapefile imports, street sign images.

S3/minio | local

GIS / shapefiles

Existing shapefiles imported and linked to streets.

PostGIS optional

offline sync — background sync when connection restored

INTEGRATION

Shapefile import

Batch upload .shp, .dbf via server script.

Python | geopandas

Tile server / basemap

OpenStreetMap / satellite for background.

leaflet

Export to GIS

Verified data → shapefile, GeoJSON for planners.

download

primary DB cluster MariaDB · photo archive file system / S3 · shape library imported layers · sync queue offline pending

Enumerator → Street verification · unit counting · photos

QC Officer → audit · validation · approve/reject

Project Manager → dashboards · installation tracking · ERP

Direction-aware left/right logic: entry direction (N/S/E/W) captured, then left/right derived automatically. Critical for house numbering consistency.

stored in Enumeration Session

DIGISTEM | built on ERPNext (Frappe)

option 2 · unified mobile UI

offline capable · realtime sync · QC workflow

You can freely copy and paste this HTML – it's a self-contained architecture diagram for DIGISTEM.